Surgical Oncology Principles

Resection Margins, Minimally Invasive Techniques, Reconstruction, and SLN Biopsy

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1. Goals of Oncological Surgery

Surgical oncology serves multiple functions across the cancer care pathway: diagnostic (excision biopsy, staging laparoscopy), curative (radical resection), cytoreductive (debulking to enhance systemic therapy efficacy), prophylactic (risk-reducing mastectomy, RRSO), and palliative (bypass, stenting, debulking for symptom control). The fundamental oncological principle is achieving adequate tumour-free margins (R0 resection) without compromising functional outcomes.

2. Resection Margin Principles

Margin Status	Definition	Clinical Impact	Tumour-Site Examples
R0	No residual tumour; all margins clear histologically	Best local control and survival	Goal in CRC, PDAC, sarcoma, NSCLC
R1	Microscopic residual disease at surgical margin	Higher local recurrence; re-excision or RT indicated	Breast (re-excision or RT); sarcoma (RT boost)
R2	Macroscopic residual disease	Palliative intent; systemic or local therapy	Debulking: ovarian, GBM, NET
CRM	Circumferential resection margin in rectal cancer	CRM ≤1 mm predicts local recurrence; neoadjuvant RT to clear	Rectal cancer: CRM >1 mm target

3. Sentinel Lymph Node (SLN) Biopsy

SLN biopsy identifies the first draining lymph node(s) from the primary tumour, allowing targeted pathological examination and sparing unnecessary lymphadenectomy. Technique: dual-agent mapping (radioisotope Tc-99m + patent blue V dye); emerging: ICG fluorescence (superior visualisation, no radioactivity). Applications:

- Breast cancer: SLN biopsy standard for clinically node-negative invasive breast cancer; ALND avoided if pN0 (SLN); ACOSOG Z0011 — ALND not required for 1-2 positive SLNs in T1-2 breast cancer undergoing BCT + whole breast RT
- Melanoma: SLN biopsy mandatory for T1b+ (>0.8 mm or ulcerated); completion lymphadenectomy no longer recommended for SLN-positive (MSLT-II, DeCOG-SLT)
- Gynaecological: SLN mapping in endometrial and cervical cancer increasingly replacing formal lymphadenectomy

4. Minimally Invasive Oncological Surgery

Procedure	Approach	Equivalence Evidence	Advantages
Laparoscopic colectomy	4-5 port laparoscopy	Non-inferior to open (COST, CLASICC)	Faster recovery, less pain, equivalent oncological outcomes
Robotic prostatectomy (RARP)	Da Vinci robotic	Equivalent to open (RCT: LAPPRO)	Improved visualisation; continence/potency preservation
VATS lobectomy	Video-assisted thoracoscopy	Non-inferior to open lobectomy	Less blood loss, shorter LOS, lower complication rate
Laparoscopic gastrectomy	4-5 port laparoscopy	Non-inferior in stage I-II (KLASS-02 — D2)	Equivalent oncological; superior QoL early
Robotic-assisted pancreatectomy	Da Vinci	Limited RCT data; centre-dependent	Distal pancreatectomy: reduced conversion; PDAC: evolving

5. Cytoreductive Surgery & HIPEC

Cytoreductive surgery (CRS) combined with Hyperthermic Intraperitoneal Chemotherapy (HIPEC) is a potentially curative approach for selected patients with peritoneal surface malignancy. Optimal candidates have Peritoneal Cancer Index (PCI) ≤20, limited disease burden, and adequate PS. HIPEC delivers heated chemotherapy (cisplatin or mitomycin C, 41-43°C, 60-90 min) directly into the peritoneal cavity at surgery, enhancing drug penetration and exploiting thermosensitivity.

- Colorectal PM: PRODIGE 7 trial (2018) questioned HIPEC benefit over CRS alone; however, ongoing debate regarding oxaliplatin-based HIPEC
- Pseudomyxoma peritonei: CRS + HIPEC is standard; 10-year OS ~70% in optimally debulked cases
- Ovarian peritoneal: OVHIPEC-1 — interval HIPEC improved OS 12 months vs. CRS alone
- Gastric PM: Limited role; select cases; NCI-sponsored trials ongoing